

2.7 Sample moments and their computation

The population quantities of this chapter have sample analogues, and the operator rules give their expectations at once. Let $\mathbf{y}_1, \dots, \mathbf{y}_n$ be $p \times 1$ observations, and stack them as the rows of the $n \times p$ data matrix $\mathbf{Y} = [\mathbf{y}_1, \dots, \mathbf{y}_n]^\top$. The script letter keeps the data matrix apart from a single random vector $\mathbf{Y}$. The rows keep the lowercase $\mathbf{y}_i$ of observed data, but in [Proposition 2.7.1](#) they are treated as random. With $\mathbf{C} = \mathbf{I}_n - n^{-1}\mathbf{1}\mathbf{1}^\top$, the sample mean vector and the **sample covariance matrix** are

$$\bar{\mathbf{y}} = \frac{1}{n} \sum_{i=1}^n \mathbf{y}_i = \frac{1}{n} \mathbf{Y}^\top \mathbf{1}, \quad \mathbf{S} = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{y}_i - \bar{\mathbf{y}})(\mathbf{y}_i - \bar{\mathbf{y}})^\top = \frac{1}{n-1} \mathbf{Y}^\top \mathbf{C} \mathbf{Y}.$$

The second form of $\mathbf{S}$ holds because $\mathbf{C}\mathbf{Y}$ is the data matrix with its column means subtracted, and $\mathbf{C}$ is symmetric and idempotent. The sample correlation matrix is $\hat{\mathbf{R}} = \hat{\mathbf{D}}^{-1}\mathbf{S}\hat{\mathbf{D}}^{-1}$, where $\hat{\mathbf{D}}$ holds the sample standard deviations.

Proposition 2.7.1. Moments of sample moments

Suppose the rows $\mathbf{y}_i$ are pairwise uncorrelated random vectors with common mean $\boldsymbol{\mu}$ and common covariance matrix $\boldsymbol{\Sigma}$. Then

- (a) $E(\bar{\mathbf{y}}) = \boldsymbol{\mu}$ and $\text{Cov}(\bar{\mathbf{y}}) = \boldsymbol{\Sigma}/n$;
- (b) $E(\mathbf{S}) = \boldsymbol{\Sigma}$;
- (c) $\mathbf{S}$ is nonnegative definite, and $\text{rank}(\mathbf{S}) \leq \min(p, n-1)$ for every realization.

Proof.

- (a) is Proposition 2.2.2(c) with $c_i = 1/n$. For (b), let $\mathbf{u}_j$ denote the j th column of $\mathbf{Y}$, the n observations of variable j . The (j, k) entry of $\mathbf{Y}^\top \mathbf{C} \mathbf{Y}$ is the bilinear form $\mathbf{u}_j^\top \mathbf{C} \mathbf{u}_k$. Because different rows are uncorrelated, $\text{Cov}(\mathbf{u}_k, \mathbf{u}_j) = \sigma_{kj} \mathbf{I}_n$, and $E(\mathbf{u}_j) = \mu_j \mathbf{1}$. By Corollary 2.3.2(d),

$$E(\mathbf{u}_j^\top \mathbf{C} \mathbf{u}_k) = \sigma_{kj} \text{tr}(\mathbf{C}) + \mu_j \mu_k \mathbf{1}^\top \mathbf{C} \mathbf{1} = (n-1) \sigma_{kj},$$

since $\text{tr}(\mathbf{C}) = n-1$ and $\mathbf{C} \mathbf{1} = \mathbf{0}$. (c) For any $\mathbf{a}$, $\mathbf{a}^\top \mathbf{S} \mathbf{a} = \|\mathbf{C} \mathbf{Y} \mathbf{a}\|^2 / (n-1) \geq 0$. The rank of $\mathbf{Y}^\top \mathbf{C} \mathbf{Y} = (\mathbf{C} \mathbf{Y})^\top (\mathbf{C} \mathbf{Y})$ equals $\text{rank}(\mathbf{C} \mathbf{Y})$, which is at most $\text{rank}(\mathbf{C}) = n-1$ and at most p . ■

The divisor $n-1$ is the trace of $\mathbf{C}$, the dimension of the space the centred data live in, exactly as $n-r$ is for regression residuals in Example (Residual sum of squares under a wrong mean). Part (c) has a practical side. With $n \leq p$, as in genomics or finance, the sample covariance matrix is always singular, even when $\boldsymbol{\Sigma}$ is positive definite. It has a linear combination with zero sample variance whether or not one exists in the population. [Chapter 28](#) returns to this.

Computation. In practice $\mathbf{C}$ is never formed: one subtracts column means. The eigenvalues of $\mathbf{S}$ are the squared singular values of $\mathbf{C} \mathbf{Y} / \sqrt{n-1}$ (Theorem 1.8.1), which can be computed without forming a cross-product matrix ([Chapter 10](#)).

Example. Macroeconomic series

Longley's data give $n = 16$ annual observations (1947–1962) on $p = 7$ US series: total employment, the GNP deflator, GNP, unemployment, armed forces, population and the year. The eigenvalues of $\mathbf{S}$ run from 9.94×10^9 down to 1.07×10^{-2} , a ratio of 9.3×10^{11} . That spread says little about the structure of the data. It mostly reflects units: the standard deviation of GNP is 99395, that of the deflator 10.79.

The correlation matrix removes the units (Proposition 2.4.1(d)). Its eigenvalues sum to $p = 7$. The largest is 5.533, so one direction carries about 79% of the standardized variation, and the smallest is 2.59×10^{-4} . By Theorem 1.7.7, the unit vector $\mathbf{v}$ belonging to the smallest eigenvalue minimizes the sample variance of $\mathbf{v}^\top \mathbf{z}$ over standardized combinations. Its two largest weights are 0.681 on year and -0.674 on GNP, and the combination has standard deviation 0.016, against 1 for each standardized variable. In these sixteen years GNP is almost an exact linear function of time (their correlation is 0.995). This is the sample version of Theorem 2.2.3(c): a nearly zero eigenvalue marks a nearly constant linear combination. For regression it signals collinearity, which Chapter 26 studies. Finally, the first 5 rows alone give a sample covariance matrix of rank 4, as Proposition 2.7.1(c) requires.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.longley.load_pandas().data
names = list(data.columns)

Y = data.to_numpy(dtype=float)           # n x p data matrix, one row per
    year
n, p = Y.shape
C = np.eye(n) - np.ones((n, n)) / n     # centring matrix I - J/n
S = Y.T @ C @ Y / (n - 1)               # sample covariance matrix
d = np.sqrt(np.diag(S))
R = S / np.outer(d, d)                  # D^{-1} S D^{-1}, D = diag(d)

lam_S = np.linalg.eigvalsh(S)
lam_R, V = np.linalg.eigh(R)             # ascending eigenvalues
print("eigenvalues of S:", lam_S)
print("eigenvalues of R:", lam_R)
v = V[:, 0]                             # direction of least variance
Z = (Y - Y.mean(0)) / d                  # standardized variables
print("sample variance of Z v:", np.var(Z @ v, ddof=1))
```

[Run this code online](#)

2.7.1 Exercises

2.7.1.1 B. Practice

Exercise. B1

Suppose the rows of $\mathbf{Y}$ are observed in time order, with common mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, and that neighbouring rows are correlated: $\text{Cov}(\mathbf{y}_i, \mathbf{y}_{i+1}) = \rho \boldsymbol{\Sigma}$, while $\text{Cov}(\mathbf{y}_i, \mathbf{y}_j) = \mathbf{0}$ for $|i - j| \geq 2$. Show that $E(\mathbf{S}) = (1 - 2\rho/n)\boldsymbol{\Sigma}$ and $\text{Cov}(\bar{\mathbf{y}}) = n^{-2}[n + 2(n - 1)\rho]\boldsymbol{\Sigma}$. Which of the two effects disappears as $n \rightarrow \infty$?

Solution. As in the proof of [Proposition 2.7.1](#), $\text{Cov}(\mathbf{u}_k, \mathbf{u}_j) = \sigma_{kj} \mathbf{T}$, where $\mathbf{T}$ is the $n \times n$ tridiagonal matrix with ones on the diagonal and ρ next to it. Since $\mathbf{C}\mathbf{1} = \mathbf{0}$, Corollary 2.3.2(d) gives $E(\mathbf{u}_j^\top \mathbf{C} \mathbf{u}_k) = \sigma_{kj} \text{tr}(\mathbf{C}\mathbf{T}) = \sigma_{kj} [\text{tr}(\mathbf{T}) - n^{-1} \mathbf{1}^\top \mathbf{T} \mathbf{1}]$. Here $\text{tr}(\mathbf{T}) = n$ and $\mathbf{1}^\top \mathbf{T} \mathbf{1} = n + 2(n-1)\rho$, so the bracket is $(n-1) - 2(n-1)\rho/n$, and dividing by $n-1$ gives $E(\mathbf{S}) = (1 - 2\rho/n)\mathbf{\Sigma}$. For the mean, $\bar{\mathbf{y}} = n^{-1} \sum_i \mathbf{y}_i$ and Proposition 2.2.2 gives n^{-2} times the sum of all $\text{Cov}(\mathbf{y}_i, \mathbf{y}_j)$: n diagonal terms $\mathbf{\Sigma}$ and $2(n-1)$ neighbouring terms $\rho\mathbf{\Sigma}$. The bias of $\mathbf{S}$ is $O(1/n)$ and disappears, but $n \text{Cov}(\bar{\mathbf{y}}) \rightarrow (1 + 2\rho)\mathbf{\Sigma}$, not $\mathbf{\Sigma}$. For $\rho > 0$ the naive standard error $\sqrt{s_{jj}/n}$ stays too small by a fixed factor however large the sample.

2.7.1.2 C. Going deeper

Exercise. C1

- With $n = 2$, show that $\mathbf{S} = \frac{1}{2} \mathbf{d} \mathbf{d}^\top$ with $\mathbf{d} = \mathbf{y}_1 - \mathbf{y}_2$. Deduce that $\text{rank}(\mathbf{S}) = 1$ whenever $\mathbf{y}_1 \neq \mathbf{y}_2$, and that every sample correlation is ± 1 , whatever $\mathbf{\Sigma}$ is.
- With $n = 3$ and $p = 3$, show that $\det(\mathbf{S}) = 0$, hence that the three sample correlations satisfy $1 + 2\hat{r}_{12}\hat{r}_{13}\hat{r}_{23} - \hat{r}_{12}^2 - \hat{r}_{13}^2 - \hat{r}_{23}^2 = 0$ when the sample variances are positive. Conclude that $\hat{r}_{23}$ is always one of the two *endpoints* of the interval allowed by Proposition 2.4.2, never an interior point.
- What does this say about reading a sample correlation matrix computed from few observations, and about the nearly singular $\hat{\mathbf{R}}$ of [Example \(Macroeconomic series\)](#)?

Solution.

- Here $\bar{\mathbf{y}} = \frac{1}{2}(\mathbf{y}_1 + \mathbf{y}_2)$, so $\mathbf{y}_1 - \bar{\mathbf{y}} = \frac{1}{2}\mathbf{d}$ and $\mathbf{y}_2 - \bar{\mathbf{y}} = -\frac{1}{2}\mathbf{d}$. Summing the two outer products and dividing by $n-1 = 1$ gives $\mathbf{S} = \frac{1}{2} \mathbf{d} \mathbf{d}^\top$, of rank one. Then $s_{jk} = d_j d_k / 2$ and $\hat{r}_{jk} = d_j d_k / \sqrt{d_j^2 d_k^2} = \text{sign}(d_j d_k)$, provided $d_j, d_k \neq 0$. The two observations determine one direction, and every pair of coordinates moves along it.
- By [Proposition 2.7.1\(c\)](#), $\text{rank}(\mathbf{S}) \leq n - 1 = 2 < 3$, so $\det(\mathbf{S}) = 0$. Writing $\mathbf{S} = \hat{\mathbf{D}} \hat{\mathbf{R}} \hat{\mathbf{D}}$ with $\hat{\mathbf{D}}$ the diagonal matrix of sample standard deviations, which is nonsingular when all sample variances are positive, gives $\det(\hat{\mathbf{R}}) = 0$. Expanding the 3×3 determinant gives the stated identity, which is a quadratic in $\hat{r}_{23}$ with roots $\hat{r}_{12}\hat{r}_{13} \pm \sqrt{(1 - \hat{r}_{12}^2)(1 - \hat{r}_{13}^2)}$, the two endpoints in Proposition 2.4.2. A population correlation matrix hits these endpoints only when one variable is an exact linear function of the other two; three observations force it.
- With $n \leq p$ the sample correlation matrix is singular whatever the population is, so its smallest eigenvalues and any statement derived from them describe the sample size as much as the data. Longley's $\hat{\mathbf{R}}$ has $n = 16$ and $p = 7$, so its near singularity is not forced by the rank bound; it is a genuine feature of the series.

Exercise. C2

Let $\mathbf{S}$ be positive definite and let $\lambda_{\min}$ be its smallest eigenvalue.

- Show that $\lambda_{\min} = \min\{\mathbf{a}^\top \mathbf{S} \mathbf{a} : \|\mathbf{a}\| = 1\}$ (Theorem 1.7.7), while $1/(\mathbf{S}^{-1})_{jj} = \min\{\mathbf{a}^\top \mathbf{S} \mathbf{a} : a_j = 1\}$ (Theorem 1.6.4).

- (b) Deduce that $\lambda_{\min} \leq 1/(\mathbf{S}^{-1})_{jj}$ for every j , with equality iff $\mathbf{e}_j$ is an eigenvector of $\mathbf{S}$ belonging to $\lambda_{\min}$.
- (c) For centred data, $\mathbf{a}^\top \mathbf{S} \mathbf{a} = \|\mathbf{C} \mathbf{y} \mathbf{a}\|^2 / (n-1)$, so $1/(\mathbf{S}^{-1})_{jj}$ is the sample variance, with divisor $n-1$, of the residuals of the least squares regression of variable j on the other $p-1$ variables, while $\lambda_{\min}$ minimizes over *all* normalized combinations. Using the smallest eigenvalue 2.59×10^{-4} of the Longley correlation matrix in [Example \(Macroeconomic series\)](#), what can you say about how well any one standardized series is predicted by the other six?

Solution.

- (a) The first statement is the Rayleigh quotient characterization. For the second, order the variables so that $j = 1$ and partition $\mathbf{a} = (1, \mathbf{b}^\top)^\top$. Then $\mathbf{a}^\top \mathbf{S} \mathbf{a} = s_{11} + 2\mathbf{b}^\top \mathbf{S}_{21} + \mathbf{b}^\top \mathbf{S}_{22} \mathbf{b}$, a positive definite quadratic in $\mathbf{b}$, minimized at $\mathbf{b} = -\mathbf{S}_{22}^{-1} \mathbf{S}_{21}$ with value the Schur complement $s_{11} - \mathbf{S}_{12} \mathbf{S}_{22}^{-1} \mathbf{S}_{21}$, which by Theorem 1.6.4 is $1/(\mathbf{S}^{-1})_{11}$.
- (b) Any $\mathbf{a}$ with $a_j = 1$ has $\|\mathbf{a}\| \geq 1$, so $\mathbf{a}^\top \mathbf{S} \mathbf{a} \geq \lambda_{\min} \|\mathbf{a}\|^2 \geq \lambda_{\min}$; minimizing over such $\mathbf{a}$ gives the inequality. Equality forces a minimizer with $\|\mathbf{a}\| = 1$ and $a_j = 1$, hence $\mathbf{a} = \mathbf{e}_j$, and $\mathbf{e}_j$ attains the Rayleigh minimum only if it is an eigenvector for $\lambda_{\min}$. Conversely, if it is, both sides equal $\lambda_{\min}$.
- (c) The residual sample variance of any one standardized series regressed on the other six is at least 2.59×10^{-4} , so by Proposition 2.5.3 its multiple correlation with the others satisfies $1 - R^2 \geq 2.59 \times 10^{-4}$. Collinearity as measured by the smallest eigenvalue is therefore always at least as severe as collinearity measured one variable at a time: the direction of least variance need not be close to any coordinate direction, and in Longley's data it is a contrast between year and GNP rather than a single series.